Soccer league - Business rules

Written for testing. Based on what the code of the soccer sample does.

What this system is for

- The system keeps the records of a soccer league: the teams, their players, the matches they play, the goals scored and the competitions the teams enter.
- League staff use it. It has no page for fans; other systems read its data through its API.

Teams

- A team has a name, a city, a country and a coach.
- Two teams cannot have the same name.
- A team can be deleted only if it has never played a match. Its players are deleted with it.
- Only an administrator can delete a team.

Players

- A player belongs to exactly one team.
- A player has a name, a shirt number, an age, a preferred foot (left, right or both) and a position (goalkeeper, defender, midfielder or forward).
- The shirt number is between 1 and 99. Two players of the same team cannot have the same number.
- A player must be at least 15 years old.
- A team cannot have more than 25 players.
- A player who is 33 or older is shown as a veteran.
- The system counts the goals of each player. An administrator can set the goals of all the players of a team back to zero, for example when a new season starts.

Matches

- A match has a home team, an away team and a stadium.

- A team cannot play against itself.
- A match starts with the score 0 - 0 and is not finished.
- The scoreboard shows the two team names and the goals of each one.
- When a match is finished it cannot be finished again.
- The winner is the team with more goals. If both have the same number of goals the match is a draw and there is no winner.

Goals

- A goal is scored in a match, by a player, at a minute.
- The minute is between 0 and 130, so extra time is allowed.
- A goal can be added only while the match is not finished.
- The team that scores must be one of the two teams of the match, and the scorer is identified by the shirt number in that team.
- Recording a goal does three things together: the player has one more goal, the score of the match changes, and the goal is saved with its minute. If one of the three fails, none of them is kept.

Competitions

- A competition has a name and a season. The season is a year with four digits.
- Two competitions cannot have the same name.
- A team can be registered in a competition only once.
- A team can be registered in several competitions.

What the system does not do

- It does not schedule matches or keep their dates.
- It does not keep standings, points or tables.
- It does not handle cards, substitutions or injuries.
- A goal cannot be removed once it is recorded.